



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Econometrics of Financial Markets

Report of the Assignment for the Final Exam

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1 Preface

All analysis are done using Gretl 2024b. Throughout the text, variables are written in *italic* exactly as they are entered the program to make following results from the appendices easier. In short, *gdp* is growth rate of GDP, *infl* is inflation rate, *rate* is policy interest rate, *m* is quantity of money, *y* is a proxy for GDP, and *i* is short-term interest rate.

2 The First Exercise

2.1 Building the Vector Autoregressive Model and Diagnostic Checks

First we need to check for stationarity of the variables. This can be done either by visual inspection (Figure 1) or by using Augmented Dickey-Fuller (ADF) test (Figure 2). We see no time-dependent trend in variables, and the ADF test rejects the null hypothesis of presence of a unit root, therefore we are safe to assume all variables are stationary.

To determine the optimal number of lags for vector autoregressive (VAR) model, we compare the information criteria for up to 24 lags (Figure 3). A single lag achieves the least value in all three criteria (AIC, BIC, HBQ), thus we will proceed with one lag for our variables.

Results of the VAR(1) are presented in Appendix B.1. Checking F-Test(3,395), the value is significant at 1% level for all equations, meaning that at least one of the coefficients are significantly different than zero in each case.

Residuals of this model are an important indicator of its quality, and Gretl offers three checks on residuals of a VAR: Autocorrelation, Normality, and Autoregressive Conditional Heteroskedasticity (ARCH). If residuals are serially correlated, the model has failed in capturing all the dynamics between variables with the provided lags. Looking at the results at Figure 4a, we see the null hypothesis of no significant autocorrelation is rejected only for 10 and 11 lags at 5%, which are irrelevant to our VAR(1). So we confidently say residuals are not autocorrelated.

We fit an ARCH model with 12 lags on residuals. The purpose is to see if heteroskedasticity is present and residuals grow with time or nor. The results in Figure 4b suggest residuals are constant over time and there is no need to consider heteroskedasticity.

Testing normality is performed by Doornik-Hansen Test, which is based on skewness and kurtosis of the residuals (Figure 4c). The null hypothesis of Doornik-Hansen test is rejected, which means residuals are not multivariate normally distributed. This undermines our ability in forecasting or inference. Even though the residuals do not exhibit autocorrelation or heteroskedasticity, we are forced to turn "Robust Standard Errors" on for the rest of the analysis to continue using statistics like t-test or F-test, all of which assume normally distributed residuals.

2.2 Investigating Granger Causality of Policy Interest Rate

A variable Grange-causes another variable if it is useful in forecasting that variable, and both variables are stationary. Note stationarity was verified in Section 2.1 before constructing the VAR. Here, the inclusion of lagged *rate* in VAR equation of *gdp* or *infl* must prove to add statistically significant information to the system that is otherwise incomplete without this term.

It should be noted since our VAR system only has 1 lag of each variable, we could perform this inspection based on individual significance of the lagged interest rate in equations alone; but we will explain the general way that Granger causality is inspected.

This can be checked by looking at the F-test of zero restrictions in VAR equations at Appendix B.1, where the null hypnosis is that these coefficients are jointly equal to zero, thus that variable adds no information to regression of the variable of interest. To clarify with an example, a restricted model for *GDP* is a regression on its own lagged terms alone, while unrestricted model also adds lagged terms of inflation and interest rate. If the coefficients on *rate* are jointly different than zero, then *rate* Granger-causes *gdp*.

F-test for all lags of *rate* rejects its null hypothesis in GDP equation at 1%, and we see coefficient for *rate*(-1) is also significant at 1%. So we can confidently say *rate* Granger-causes *gdp*.

Conversely, F-test of interest rate does not reject the null hypothesis at all for *infl*, and in fact we see coefficient on *rate*(-1) is not statistically different than zero. Therefore, *rate* does not Granger-cause *infl*.

2.3 Monetary Transmission Mechanism and Impulse Response Functions

One limitation of VAR is that it is not suitable for economic interpretations of finding cause and effect. To solve this issue, Gretl offers Cholesky decomposition approach to build a structural vector autoregressive (sVAR) model, which can examine the propagation of a shock in our system of variables. Order of variables in a Cholesky decomposition shows their impact on the other ones; meaning that the first variable is not affected contemporaneously by others, while the last variable does not affect any of the variables before contemporaneously.

Based on economic insights, we speculate the order from the first to the third is: *gdp*, *infl*, and *rate*. The reasoning is that GDP - an indicator of economic activity - is slower to change, while any shocks to economic

activity could pose immediate impact on interest rate and inflation. Inflation rate can have a contemporaneous effect on interest rate, and in fact, the best tool that central banks have at disposal to fight inflation is the interest rate. The resulting plots using bootstrapping technique to have a confidence interval of $1 - \alpha = 0.68$ are given in Figure 5, with the numerical values in Appendix B.2.

A positive shock to interest rate does not immediately change the growth rate of GDP or inflation rate, but it reduces both variables in the short run. Growth rate of GDP seems to be more resilient to changes in interest rate; since the negative impact stops after 3 months, and almost dissipates after 11 months (0.0004). Inflation rate sustains a downward trend for a bit longer at 4 months, but reaches equilibrium slightly faster (0.0004 after 10 months).

Looking at the effect of GDP or inflation shock on interest rate, we see **Taylor Rule** in effect. This rule states that federal banks should increase interest rate when either inflation or GDP growth are above target levels. This is indeed the case here, where we see a positive shock on GDP induces an immediate raise to interest rate, which will dissipate over 10 months. A positive shock on inflation rate sharply increases the interest rate for 4 months, after which the interest rate declines along with inflation rate itself.

One other noticeable impact is the **Pigou Effect**, which states when there is a period of deflation, companies hire more workers to increase production, which will eventually lead to more consumption (Growth of GDP). Here we see the reverse of this effect; which means in response to a positive shock to Inflation rate, growth rate of GDP decreases in the short run. In fact, after the shock changes to a negative effect on Inflation after 3 months, we see a corresponding reversal in change of GDP growth trajectory.

2.4 Comparing Autoregressive Moving Average (ARMA) Model with Vector Autoregressive (AR) Model

ARMA(p,q) is a model that simultaneously considers the effect of previous assumed values of the variable via AR(p) and past error terms using MA(q). For this section, we have chosen to model **Inflation Rate** with ARMA and compare it with the VAR(1) equation from Section 2.1.

First, we need to identify the optimum order for autoregressive and moving average legs of the ARMA. This can be done by looking at the correlogram of *rate* in Figure 6. The purpose of AutoCorrelation Function (ACF) is to show serial autocorrelation in time series. Partial AutoCorrelation Function (PACF) does the same, but also controls for the correlation between the values in the interval up to that lag. Based on the values in Table 6a, we can choose 2 lags because there are two spikes at the ACF, followed by mostly insignificant ones for the rest.

We can also find the optimum number of lags from minimization of information ratio. Figure 7 shows an ARMA(2,1) achieves the minimum value for all three criteria, which is in line with our findings from correlogram in Figure 6.

The resulting ARMA(2,1) can be found in Appendix B.3, where coefficients for all parameters are significantly different than zero. Trying ARMA(2,2) results in θ_2 being statistically insignificant. Regarding parsimony of the model, reducing ARMA legs results in even worse performance compared to VAR, with higher information criteria across the board. Therefore, ARMA(2,1) is the best specification for *rate*.

The residuals of this model follow a Normal distribution (Figure 8). Ljung-Box test also rejects the null hypotheses of presence of autocorrelation between the residuals (Figure 9a). Fitting an ARCH model on the residuals concludes that no ARCH effect is present, which means there is no heteroskedasticity in error terms (Figure 9b).

First thing to notice is that R^2 is lower in ARMA model compared to VAR. This is natural, since the VAR has more regressors. However, even $R^2 - adjusted$ which penalizes the number of regressors, is higher for VAR (0.3862 versus 0.3405), meaning VAR can explain more of the variance in *rate*. Looking at the VAR model of *rate*, we can attribute its better performance to inclusion of a statistically significant lagged term of the GDP Growth - something that ARMA lacks. This also verifies with economic insights; shocks to GDP can and will influence inflation rate. Since the VAR model considers the interrelationships between variables, while the ARMA model only models the variable itself, this can explain why VAR might be better to model inflation rate in our case.

3 The Second Exercise

3.1 Checking for the Stationarity of the Variables

Similar to Section 2.1, stationarity can be checked either by visual inspection of the time series (Figure 10) or performing an ADF test (Figure 11). Visually, we can see a linear trend in *m*, and *y*, while *i* might follow a stochastic trend. ADF test also firmly rejects the null hypothesis of stationarity for all three of the variables, therefore we can say our variables are non-stationary in levels.

We check the first difference of variables to see if they are integrated of order 1. Time plots in Figure 12 along with ADF tests in Figure 13 prove this is the case.

3.2 Checking for Cointegration

We established our variables are non-stationary in levels, therefore the money demand equation must be a cointegrating regression meaning that even though individual variables are non-stationary, their linear combination is stationary. If this is not the case, then any deviation from equilibrium would be permanent. To assess this behavior, Gretl offers two tests of cointegration tests: Engle-Granger or Johansen. The latter allows for inspecting more than one cointegrating linear regression, but we are asked to use the former test. The rest of this Section along with Section 3.3 will follow this methodology, while Section 3.4 uses Johansen test and VECM to give a more generalized answer to the assignment question.

By setting m as the dependent variable, Engle-Granger test results in Figure 14. With a very low p -value, we strongly reject the null hypothesis of presence of a unit root in regression residuals, which means the residuals are stationary. Combined with non-stationarity of the individual variables from Section 3.1, we conclude quantity of money equation is indeed a cointegrating regression.

On a side note, we notice the coefficient of y in the cointegrating regression is positive (1.0041), while the coefficient of i is negative (-0.2859), and both coefficients are strongly significant. This verifies the economic theory the given equation.

3.3 Error Correction Model (ECM) and Discussion on m Reacting to Disequilibria

We established these variables are cointegrated, so when a disequilibria happens, there must be an error correction force that drives equilibrium error back to zero and guarantees stationarity. We can model this as:

$$\Delta m_t = \phi_0 + \phi_1 \Delta m_{t-1} + \phi_2 \Delta y_{t-1} + \phi_3 \Delta i_{t-1} + \alpha \times \hat{u}_{t-1} + \epsilon_t$$

where $\hat{u}_t$ are the residuals from OLS regression at time t on the level of variables. The resulting equation is called an Error Correction Model (ECM). The residuals from the cointegrating regression (Appendix B.4) are saved and put in the ECM. The results can be found in Appendix B.5.

Notice the coefficient on $\Delta m_{(-1)}$ is not statistically significant, implying past short-term changes in quantity of money have no persistence. Perhaps more importantly, the coefficients on $\Delta y_{(-1)}$ and $\Delta i_{(-1)}$ are not statistically significant either. This shows neither GDP nor short-term interest rate affects quantity of money in the short run when a disequilibrium happens. On the other hand, coefficient on lagged OLS residuals is significant at 1% and has a negative value of -0.3691. This coefficient shows the speed of error correction (adjustment rate) and is expected to be negative to be able to pull back the equation to long-term equilibrium, which is the case here.

From economic point of view, if money supplies are higher than the long-run equilibrium, then approximately 37% of the excess (disequilibria) from time $t - 1$ will be decreased (corrected) at time t and so on. The conclusion is that quantity of money reacts and adjusts itself when a disequilibrium happens.

3.4 Vector Error Correction Model (VECM) and Discussion on All Variables Reacting to Disequilibria

The procedure in Sections 3.2 & 3.3 are only applicable when we are interested in adjustments to a single variable. If we want to answer the more general question that "which variables react to disequilibria?" we need a more sophisticated method.

First we apply Johansen test to determine the rank of cointegration (Figure 15). Trace test states there are at least as many cointegrating equations as Rank. This statistic is rejected for rank 0, but not rank 1, meaning we have only 1 cointegrating equation - which we are already given in the assignment. Also notice the results for α matrix, which shows adjustment vectors. From the first column, we see this value is large for m , small for i , and almost zero for y . This makes us expect only m and i to react meaningfully to disequilibria; but we should construct Vector Error Correction Model (VECM) to reach a proper conclusion.

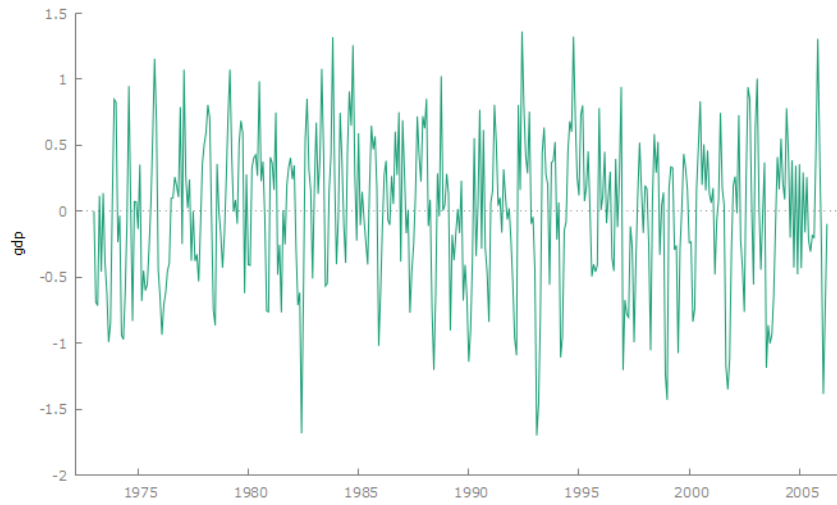
Since the optimum number of lags for corresponding VAR is 1, we deduce 1 from it to get zero lagged variables. Consequently, the VECM will only have first difference of each variable regressed on a constant term and the 1-lag error correction. The results for VECM system are provided in Appendix B.6. Notice for the Δm , we got a statistically significant lagged error correction ($EC1$) term, with a coefficient similar to what we deduced from ECM based on Engle-Granger in Section 3.3 (-0.3575, or approximately 36% error correction per period).

Looking at other equations, Δy has a fairly high p -value for $EC1$, thus it is not statistically significant under our usual margins. This means GDP reacts almost as an exogenous variable in a disequilibria and does not adjust to accommodate the change in the short run. On the other hand, Δi has a statistically significant, albeit at 5% level, coefficient for $EC1$. This coefficient is positive, which means when there is a disequilibria, short-time interest rate increases in response to a positive disequilibria. Also notice the coefficients for error correction terms are the same as adjustment vectors from Johansen test, but this time we also have the t-statistic and can see their significance.

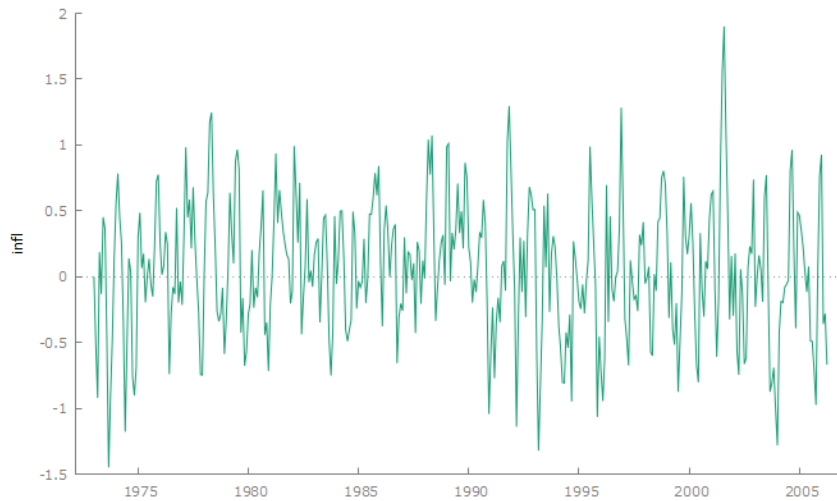
In conclusion, we can say only money demand and interest rate react to a disequilibria in the cointegrating equation, and the role of i latter is weaker and slower compared to m .

A Graphs and Tables for Tests

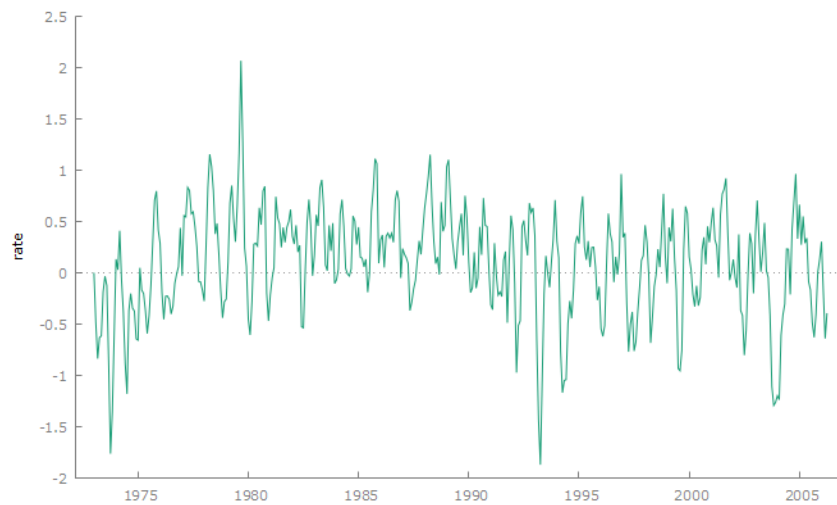
- Exercise 1 -



(a) Growth rate of the GDP



(b) Inflation rate



(c) Policy interest rate

Figure 1: Time plot of the three variables of the first exercise

```

gretl: ADF test
Augmented Dickey-Fuller test for gdp
testing down from 16 lags, criterion AIC
sample size 396
unit-root null hypothesis: a = 1

test with constant
including 3 lags of (1-L)gdp
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.778468
test statistic: tau_c(1) = -11.0914
asymptotic p-value 9.655e-23
1st-order autocorrelation coeff. for e: 0.003
lagged differences: F(3, 391) = 6.641 [0.0002]

with constant and trend
including 3 lags of (1-L)gdp
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.779213
test statistic: tau_ct(1) = -11.0867
asymptotic p-value 2.418e-24
1st-order autocorrelation coeff. for e: 0.003
lagged differences: F(3, 390) = 6.631 [0.0002]

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(a) ADF test for GDP

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gretl: ADF test
Augmented Dickey-Fuller test for infl
testing down from 16 lags, criterion AIC
sample size 397
unit-root null hypothesis: a = 1

test with constant
including 2 lags of (1-L)infl
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.572943
test statistic: tau_c(1) = -10.8285
asymptotic p-value 6.944e-22
1st-order autocorrelation coeff. for e: 0.001
lagged differences: F(2, 393) = 8.106 [0.0004]

with constant and trend
including 2 lags of (1-L)infl
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.575785
test statistic: tau_ct(1) = -10.8655
asymptotic p-value 2.169e-23
1st-order autocorrelation coeff. for e: 0.000
lagged differences: F(2, 392) = 8.204 [0.0003]

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(b) ADF test for Infl

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gretl: ADF test
Augmented Dickey-Fuller test for rate
testing down from 16 lags, criterion AIC
sample size 384
unit-root null hypothesis: a = 1

test with constant
including 15 lags of (1-L)rate
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.224081
test statistic: tau_c(1) = -3.3743
asymptotic p-value 0.0119
1st-order autocorrelation coeff. for e: -0.007
lagged differences: F(15, 367) = 4.015 [0.0000]

with constant and trend
including 15 lags of (1-L)rate
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.247594
test statistic: tau_ct(1) = -3.6464
asymptotic p-value 0.02604
1st-order autocorrelation coeff. for e: -0.008
lagged differences: F(15, 366) = 4.062 [0.0000]

```

(c) ADF test for Rate

Figure 2: Augmented Dickey-Fuller (ADF) test for the three variables of the first exercise

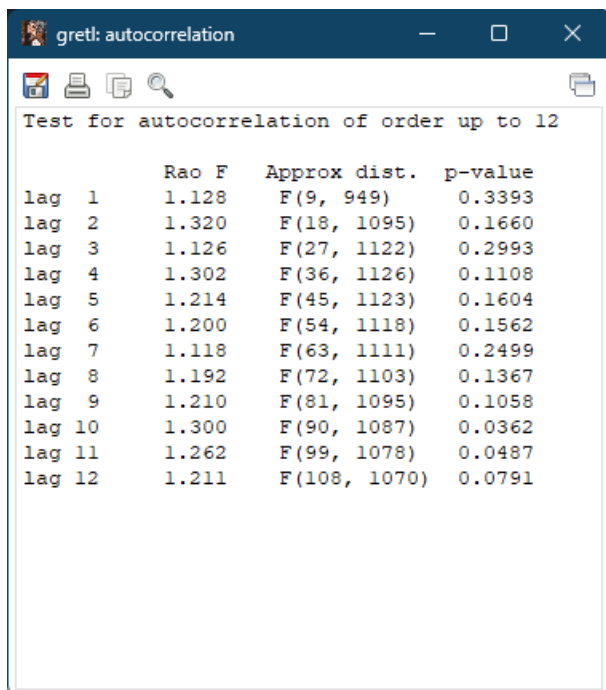
gretl: VAR lag selection

VAR system, maximum lag order 24

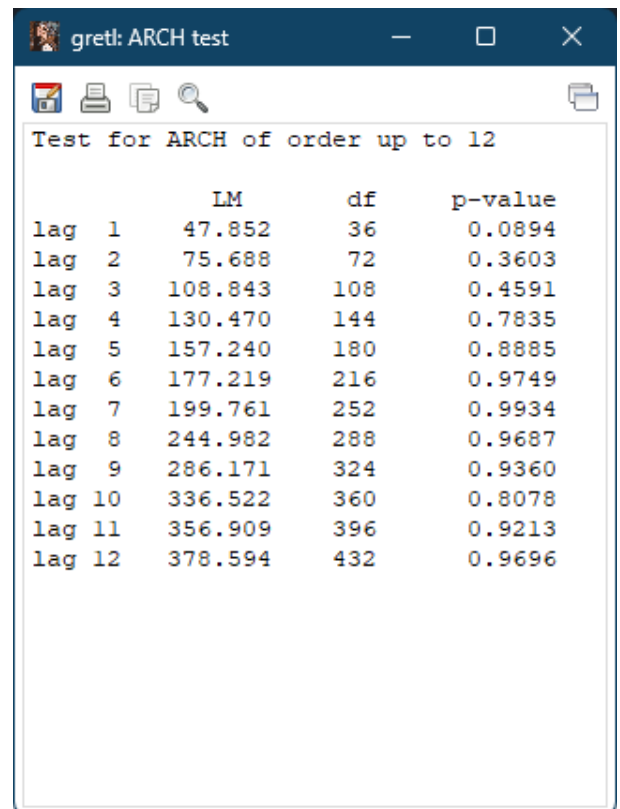
The asterisks below indicate the best (that is, minimized) values of the respective information criteria, AIC = Akaike criterion, BIC = Schwarz Bayesian criterion and HQC = Hannan-Quinn criterion.

lags	loglik	p(LR)	AIC	BIC	HQC
1	-158.98002		0.909468*	1.034881*	0.959253*
2	-152.59646	0.17343	0.923385	1.142857	1.010508
3	-146.80977	0.23844	0.940477	1.254009	1.064938
4	-144.14851	0.80534	0.974194	1.381785	1.135993
5	-135.44984	0.04285	0.975797	1.477447	1.174934
6	-131.99068	0.64562	1.005270	1.600979	1.241745
7	-126.70581	0.30636	1.025031	1.714799	1.298845
8	-122.75958	0.54502	1.051913	1.835740	1.363065
9	-114.77354	0.06747	1.057306	1.935193	1.405797
10	-105.73701	0.03434	1.057112	2.029058	1.442941
11	-98.57749	0.11142	1.066902	2.132907	1.490069
12	-96.25706	0.86442	1.102431	2.262496	1.562936
13	-92.12259	0.50729	1.128312	2.382436	1.626155
14	-88.30485	0.57126	1.155877	2.504060	1.691059
15	-81.94831	0.17602	1.169938	2.612181	1.742458
16	-74.57036	0.09786	1.178566	2.714868	1.788424
17	-70.07986	0.43903	1.202552	2.832914	1.849749
18	-66.30162	0.57938	1.230328	2.954749	1.914863
19	-60.40811	0.22558	1.246852	3.065332	1.968725
20	-58.57631	0.93213	1.284980	3.197520	2.044192
21	-50.64551	0.06983	1.290668	3.297266	2.087217
22	-47.67769	0.74634	1.322754	3.423412	2.156642
23	-46.05920	0.95415	1.362017	3.556734	2.233243
24	-40.90890	0.32670	1.382494	3.671271	2.291059

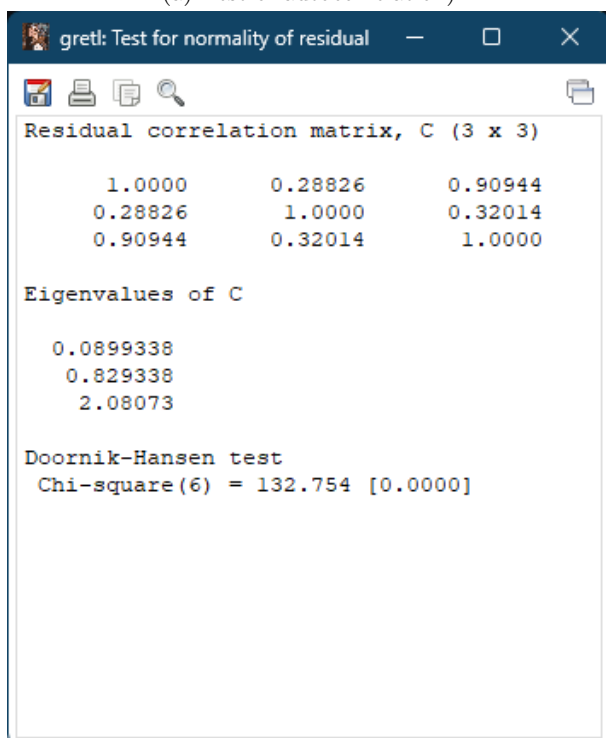
Figure 3: Vector Autoregression (VAR) optimal lag selection based on three information criteria



(a) Test of autocorrelation)



(b) Test of heteroskedasticity



(c) Test of normality

Figure 4: Diagnostic tests on residuals of the VAR(1)

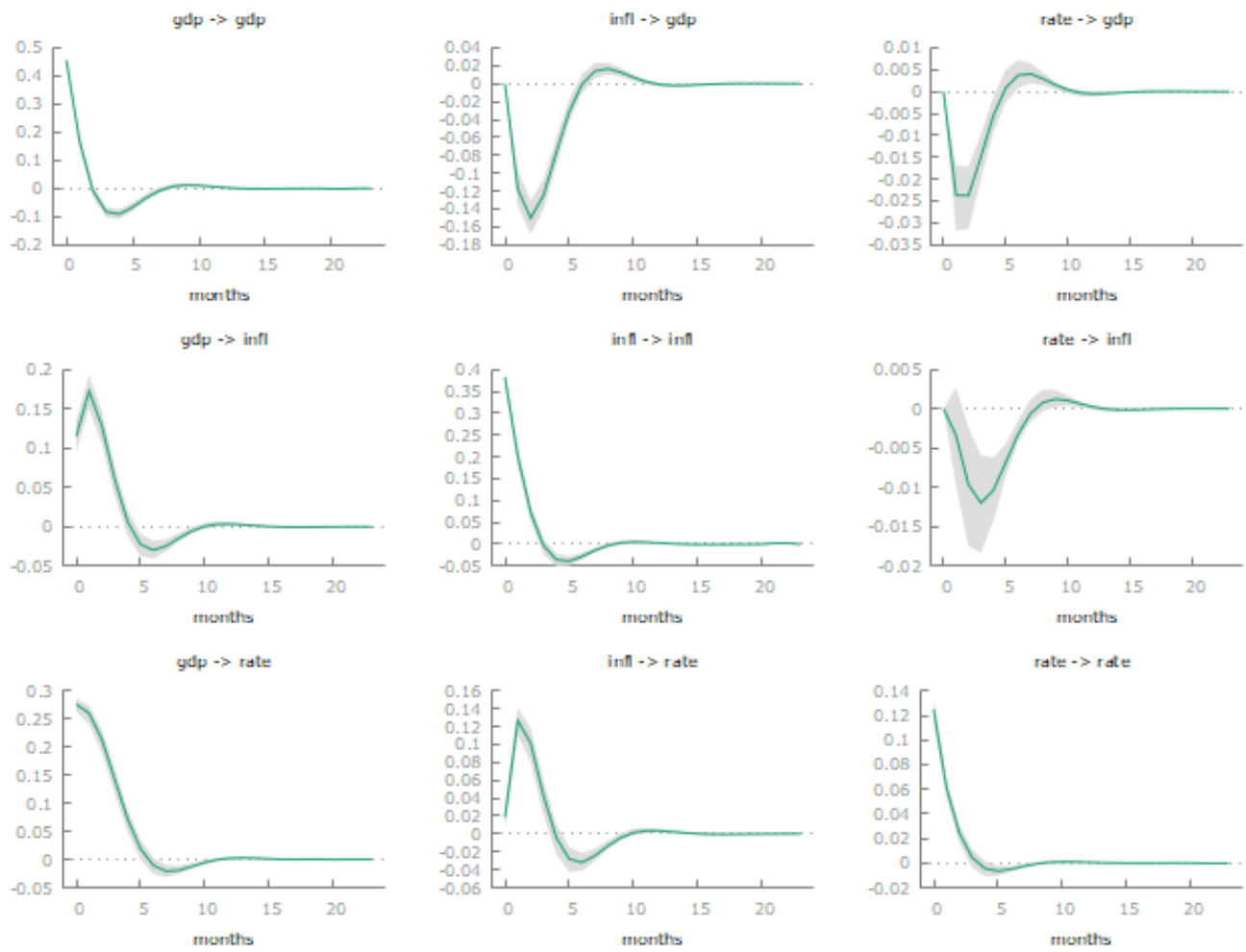
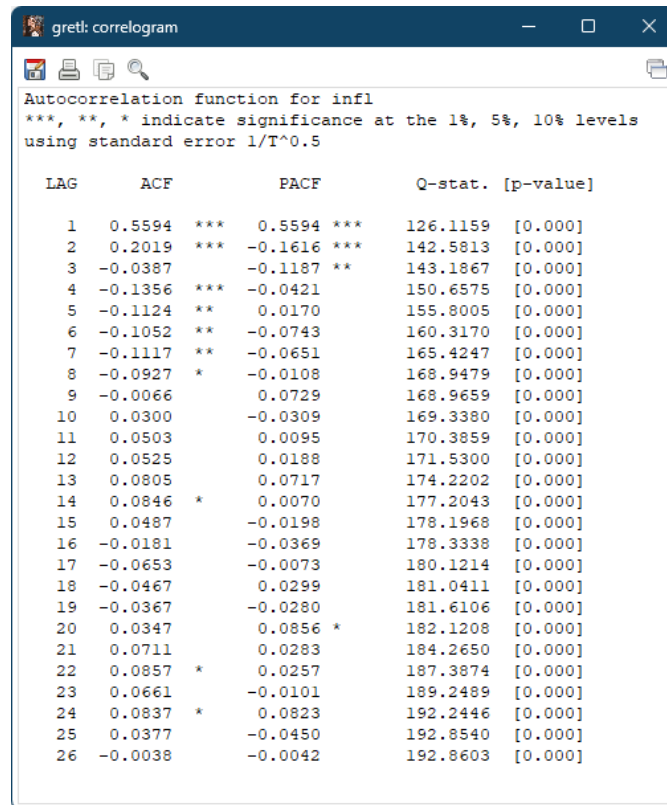
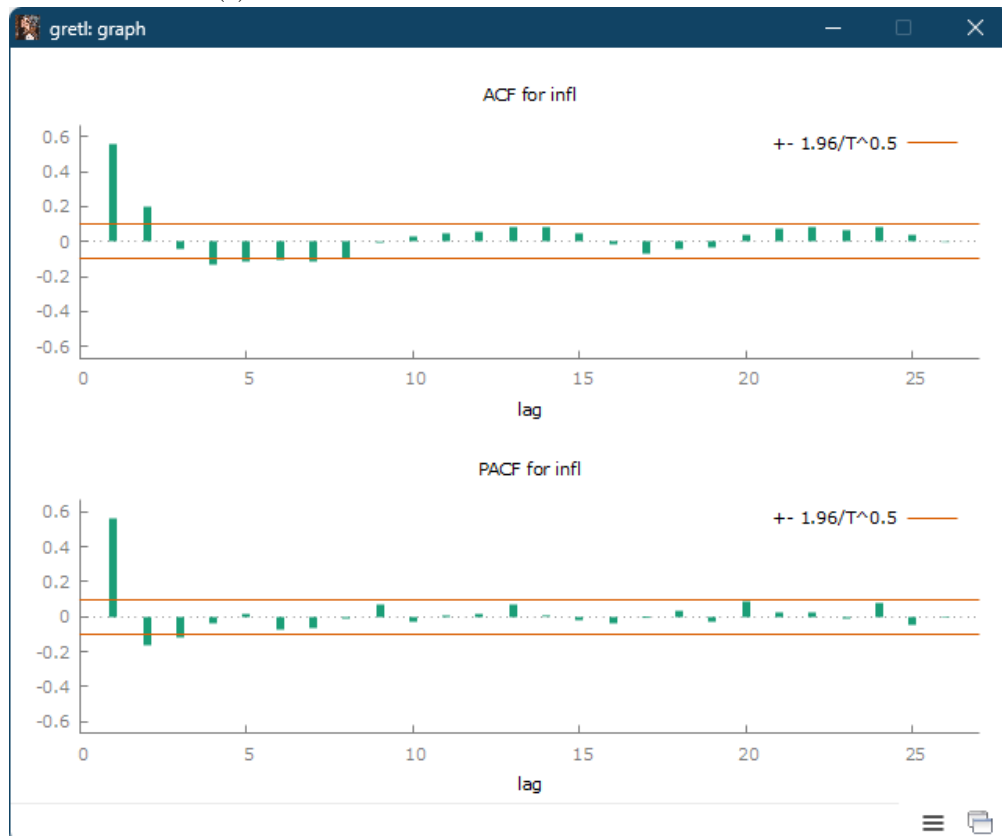


Figure 5: Impulse response function for a unit standard deviation shock in variables ($1 - \alpha = 0.68$)



(a) Table of autocorrelation function for Inflation Rate



(b) Graphical representation of autocorrelation function for Inflation Rate

Figure 6: Correlogram of the Inflation Rate

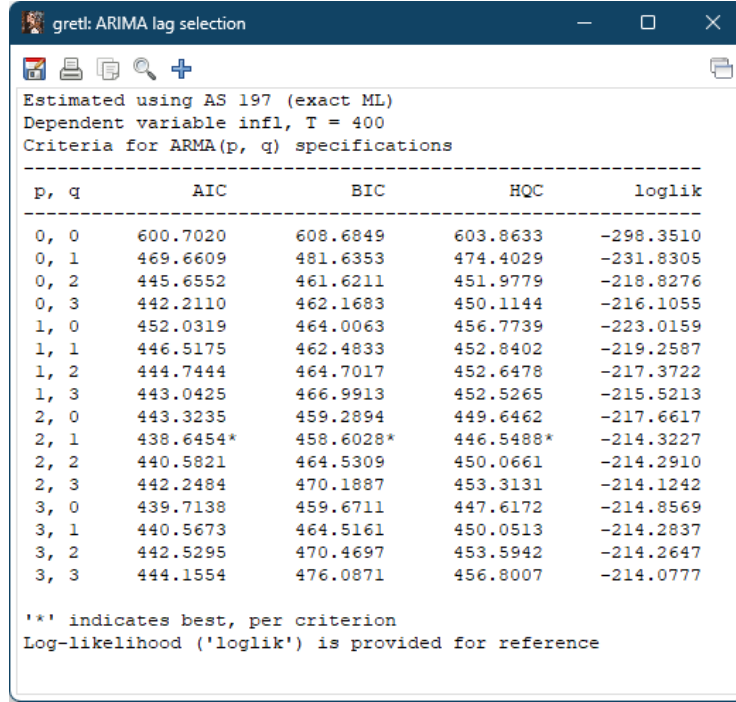
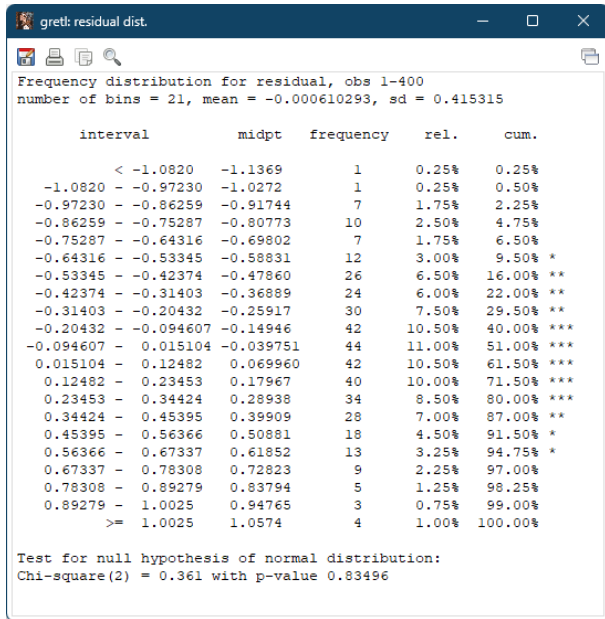
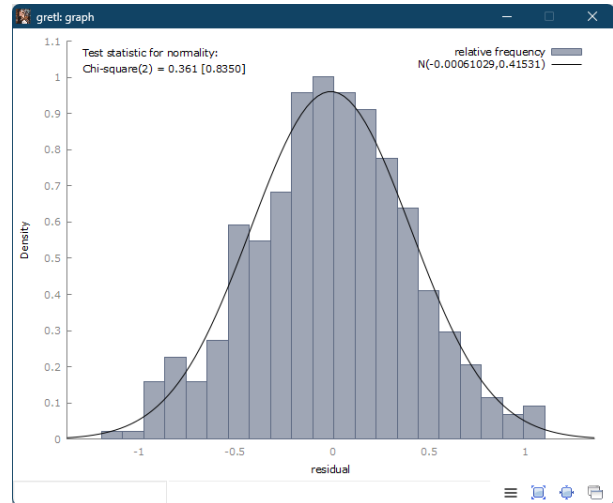


Figure 7: Information criteria for different number of lags in an ARMA(p,q) of the Inflation Rate

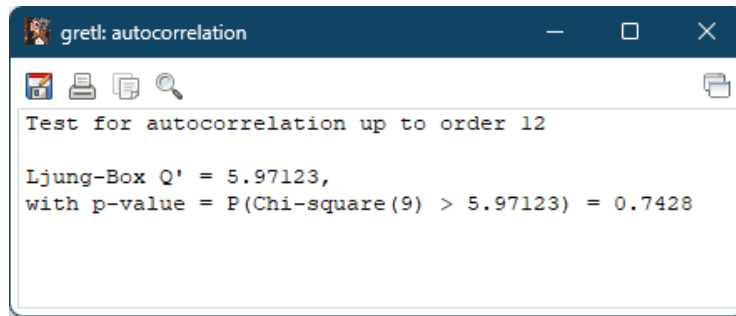


(a) Distribution of the residuals

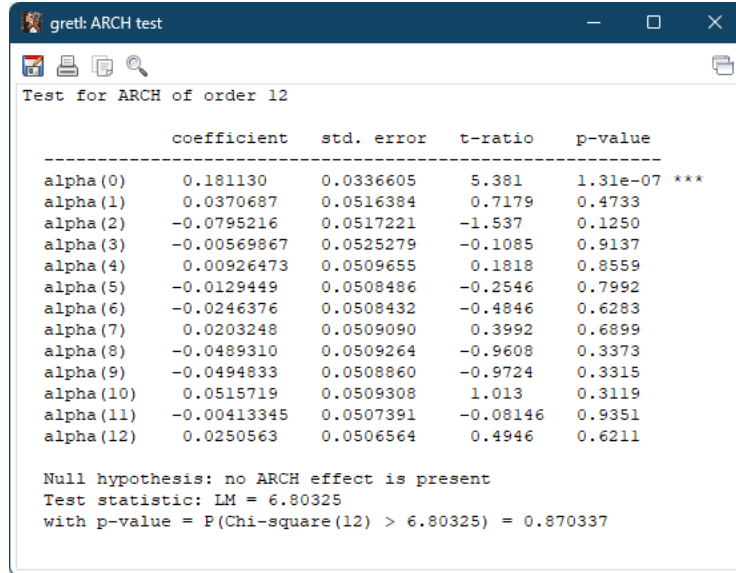


(b) Plot of the residuals

Figure 8: Test of normality for residuals of ARMA(2,1) fit on Inflation Rate



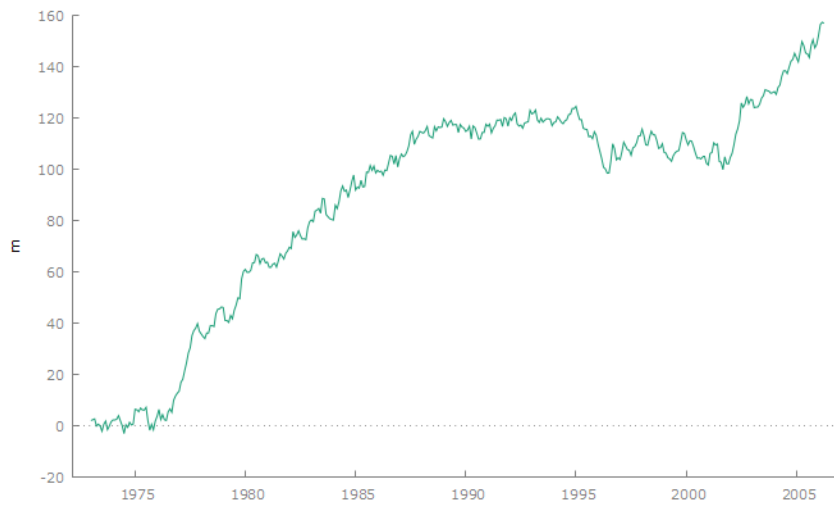
(a) Ljung-Box test of autocorrelation



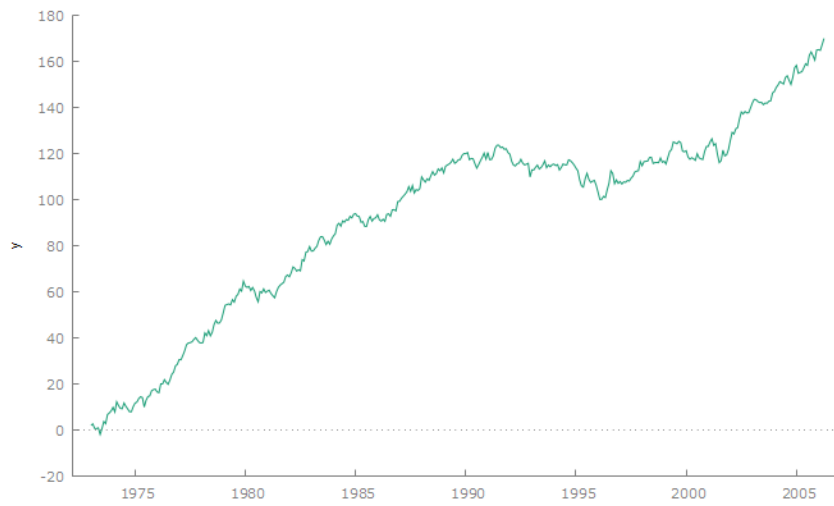
(b) Check for presence of heteroskedasticity

Figure 9: Diagnostic tests on error terms of ARMA(2,1) on Inflation Rate

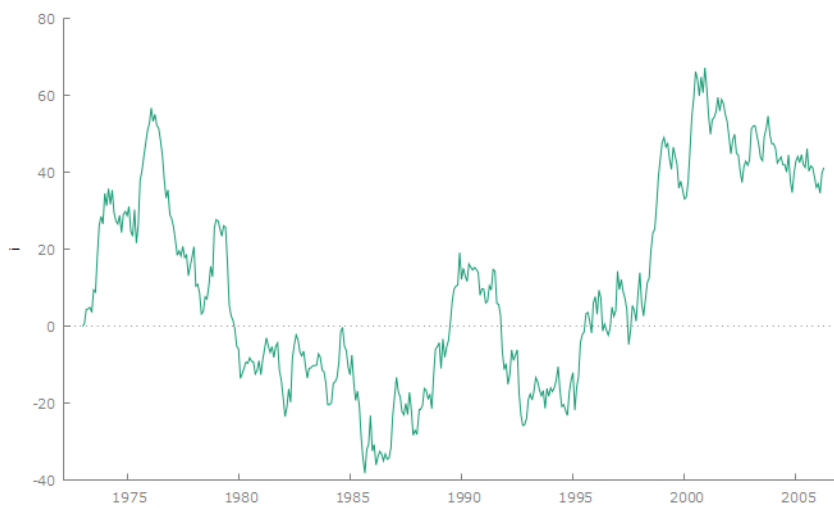
- Exercise 2 -



(a) Quantity of money (m)



(b) Gross Domestic Product (y)



(c) Short-term interest rate (i)

Figure 10: Time plot of the three variables of the second exercise

```
gretl: cointegration test

Step 1: testing for a unit root in m

Augmented Dickey-Fuller test for m
including one lag of (1-L)m
sample size 398
unit-root null hypothesis: a = 1

test with constant
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.00379927
test statistic: tau_c(1) = -1.34475
asymptotic p-value 0.6108
1st-order autocorrelation coeff. for e: 0.000

Step 2: testing for a unit root in y

Augmented Dickey-Fuller test for y
including one lag of (1-L)y
sample size 398
unit-root null hypothesis: a = 1

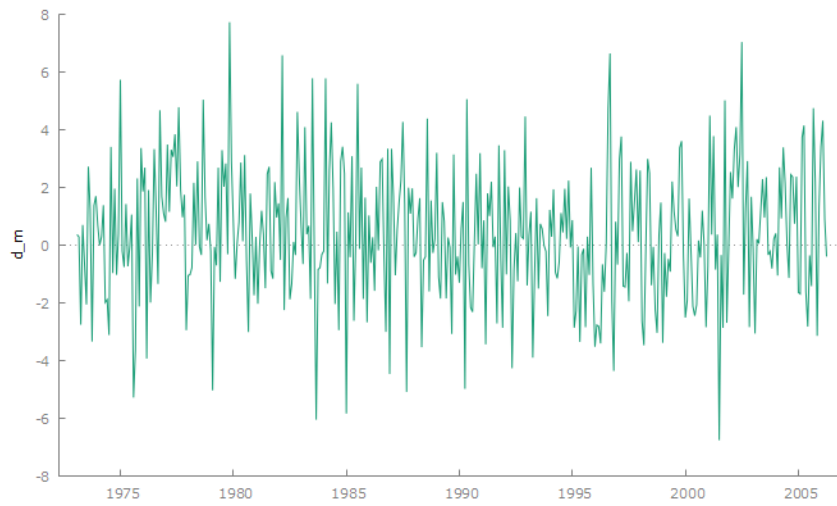
test with constant
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.00272005
test statistic: tau_c(1) = -1.21248
asymptotic p-value 0.6715
1st-order autocorrelation coeff. for e: 0.000

Step 3: testing for a unit root in i

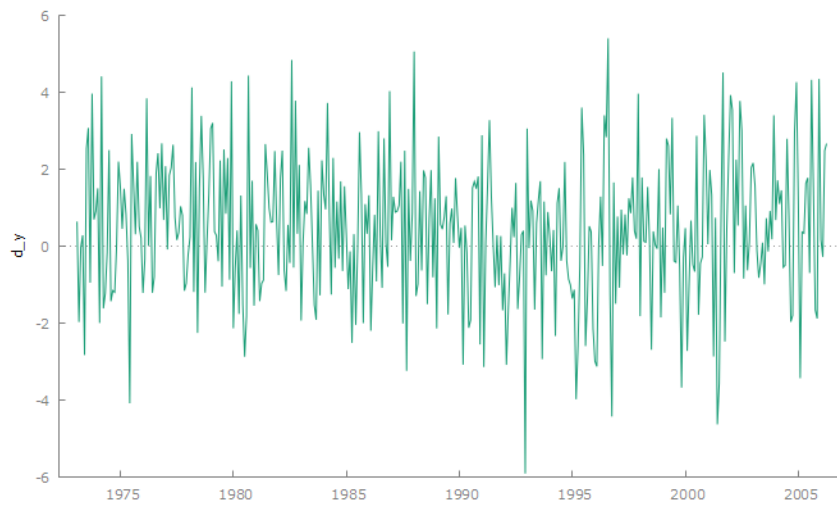
Augmented Dickey-Fuller test for i
including one lag of (1-L)i
sample size 398
unit-root null hypothesis: a = 1

test with constant
model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.01169
test statistic: tau_c(1) = -1.47392
asymptotic p-value 0.5471
1st-order autocorrelation coeff. for e: -0.004
```

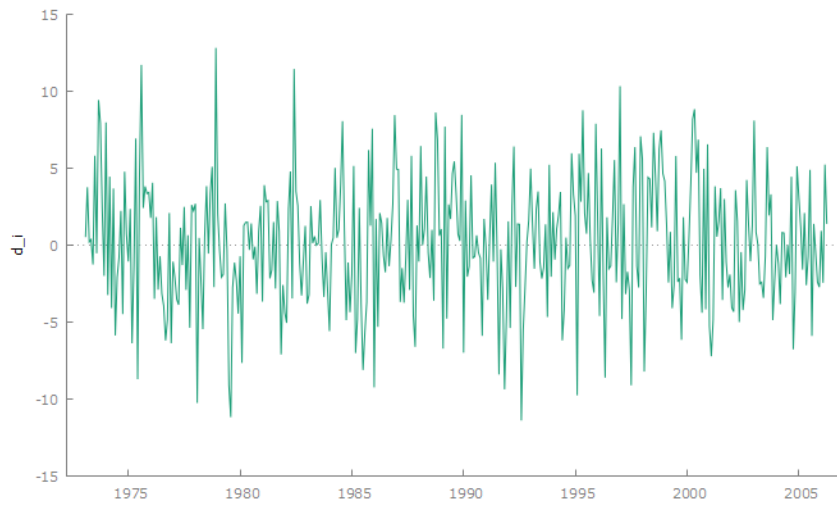
Figure 11: Augmented Dickey-Fuller Tests for Variables



(a) First order difference of quantity of money (d_m)



(b) First order difference of GDP (d_y)



(c) First order difference of interest rate (d_i)

Figure 12: Time plot of the first order difference of variables of the second exercise

```

gretl: ADF test
Augmented Dickey-Fuller test for d_m
testing down from 16 lags, criterion AIC
sample size 398
unit-root null hypothesis: a = 1

test with constant
including 0 lags of (1-L)d_m
model: (1-L)y = b0 + (a-1)*y(-1) + e
estimated value of (a - 1): -1.01527
test statistic: tau_c(1) = -20.2032
asymptotic p-value 9.504e-48
1st-order autocorrelation coeff. for e: 0.000

with constant and trend
including 0 lags of (1-L)d_m
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + e
estimated value of (a - 1): -1.01657
test statistic: tau_ct(1) = -20.205
asymptotic p-value 1.747e-70
1st-order autocorrelation coeff. for e: 0.000

```

(a) ADF test for d_m

```

gretl: ADF test
Augmented Dickey-Fuller test for d_y
testing down from 16 lags, criterion AIC
sample size 398
unit-root null hypothesis: a = 1

test with constant
including 0 lags of (1-L)d_y
model: (1-L)y = b0 + (a-1)*y(-1) + e
estimated value of (a - 1): -1.03507
test statistic: tau_c(1) = -20.5719
asymptotic p-value 2.297e-48
1st-order autocorrelation coeff. for e: 0.001

with constant and trend
including 0 lags of (1-L)d_y
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + e
estimated value of (a - 1): -1.03635
test statistic: tau_ct(1) = -20.568
asymptotic p-value 2.083e-72
1st-order autocorrelation coeff. for e: 0.001

```

(b) ADF test for d_y

```

gretl: ADF test
Augmented Dickey-Fuller test for d_i
testing down from 16 lags, criterion AIC
sample size 396
unit-root null hypothesis: a = 1

test with constant
including 2 lags of (1-L)d_i
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.808732
test statistic: tau_c(1) = -9.79678
asymptotic p-value 1.483e-18
1st-order autocorrelation coeff. for e: 0.003
lagged differences: F(2, 392) = 2.512 [0.0824]

with constant and trend
including 2 lags of (1-L)d_i
model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
estimated value of (a - 1): -0.809822
test statistic: tau_ct(1) = -9.79539
asymptotic p-value 5.574e-19
1st-order autocorrelation coeff. for e: 0.003
lagged differences: F(2, 391) = 2.486 [0.0845]

```

(c) ADF test for d_i

Figure 13: Augmented Dickey-Fuller (ADF) test for the first difference of the variables

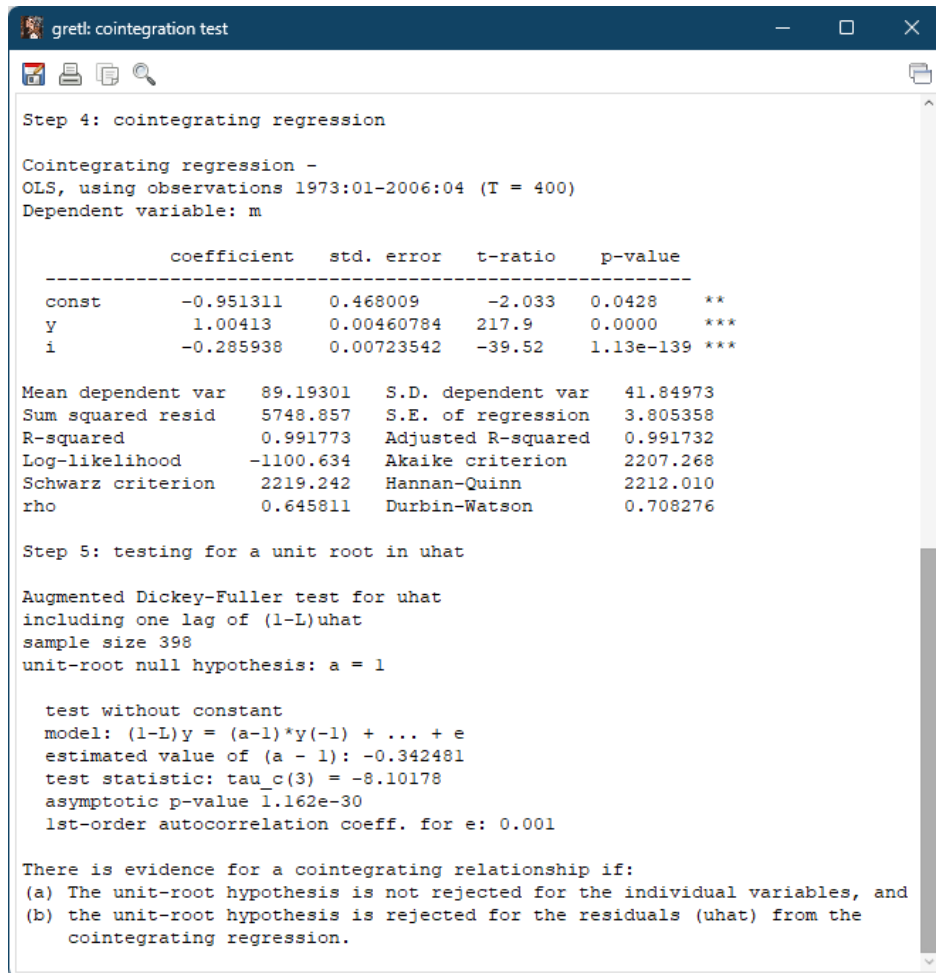


Figure 14: Engle-Granger cointegrating regression

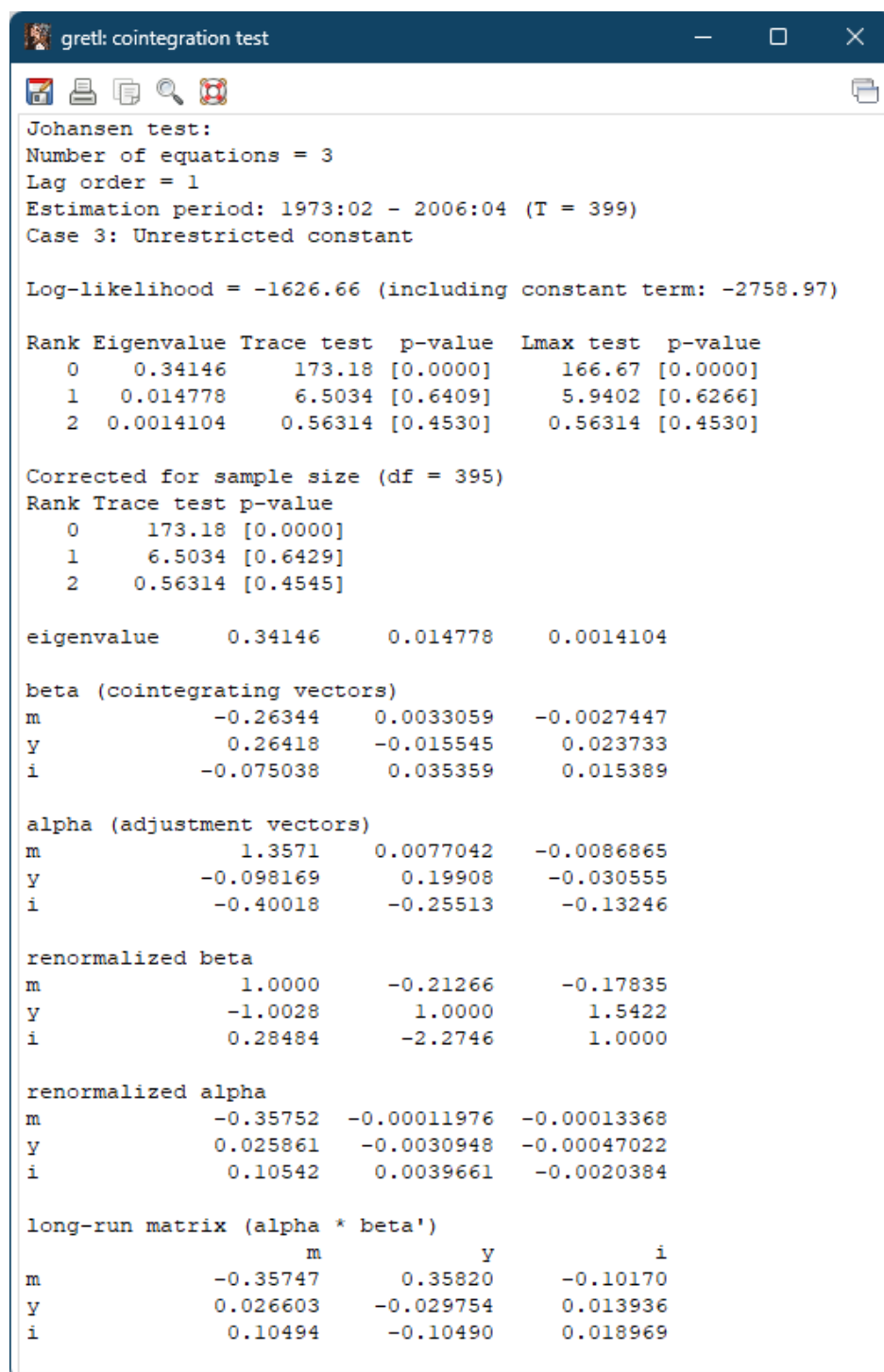


Figure 15: Johansen Test for cointegration

B Gretl Outputs for Models

- Exercise 1 -

B.1 VAR Model Specifications

VAR system, lag order 1

OLS estimates, observations 1973:02–2006:04 ($T = 399$)

Log-likelihood = -171.168

Determinant of covariance matrix = 0.000473365

AIC = 0.9181

BIC = 1.0381

HQC = 0.9657

Portmanteau test: $LB(48) = 450.067$, $df = 423$ [0.1751]

Equation 1: gdp

Heteroskedasticity-robust standard errors, variant HC1

	Coefficient	Std. Error	t -ratio	p-value
const	0.0319365	0.0230290	1.387	0.1663
gdp_{t-1}	0.557333	0.0430484	12.95	0.0000
$infl_{t-1}$	-0.299575	0.0622491	-4.813	0.0000
$rate_{t-1}$	-0.189711	0.0579095	-3.276	0.0011
Mean dependent var	-0.004411	S.D. dependent var		0.564572
Sum squared resid	83.08057	S.E. of regression		0.458618
R^2	0.345096	Adjusted R^2		0.340122
$F(3, 395)$	71.88245	P-value(F)		4.09e-37
$\hat{\rho}$	-0.035848	Durbin-Watson		2.065350

F-tests of zero restrictions

All lags of gdp	$F(1, 395) = 167.616$	[0.0000]
All lags of infl	$F(1, 395) = 23.1603$	[0.0000]
All lags of rate	$F(1, 395) = 10.7321$	[0.0011]
All vars, lag 1	$F(3, 395) = 71.8825$	[0.0000]

Equation 2: infl

Heteroskedasticity-robust standard errors, variant HC1

	Coefficient	Std. Error	t -ratio	p-value
const	0.0266110	0.0204598	1.301	0.1941
gdp_{t-1}	0.260452	0.0374016	6.964	0.0000
$infl_{t-1}$	0.530901	0.0481655	11.02	0.0000
$rate_{t-1}$	-0.0271433	0.0507275	-0.5351	0.5929
Mean dependent var	0.050719	S.D. dependent var		0.511418
Sum squared resid	63.40686	S.E. of regression		0.400654
R^2	0.390883	Adjusted R^2		0.386257
$F(3, 395)$	80.09125	P-value(F)		1.72e-40
$\hat{\rho}$	-0.001486	Durbin-Watson		1.996206

F-tests of zero restrictions

All lags of gdp	$F(1, 395) = 48.4926$	[0.0000]
All lags of infl	$F(1, 395) = 121.494$	[0.0000]
All lags of rate	$F(1, 395) = 0.286311$	[0.5929]
All vars, lag 1	$F(3, 395) = 80.0913$	[0.0000]

Equation 3: rate

Heteroskedasticity-robust standard errors, variant HC1

	Coefficient	Std. Error	<i>t</i> -ratio	p-value
const	0.0329579	0.0152464	2.162	0.0312
gdp _{<i>t</i>-1}	0.196005	0.0291425	6.726	0.0000
infl _{<i>t</i>-1}	0.308704	0.0370629	8.329	0.0000
rate _{<i>t</i>-1}	0.489711	0.0370813	13.21	0.0000
Mean dependent var	0.095627	S.D. dependent var		0.517247
Sum squared resid	36.77979	S.E. of regression		0.305145
<i>R</i> ²	0.654593	Adjusted <i>R</i> ²		0.651970
<i>F</i> (3, 395)	212.7672	P-value(<i>F</i>)		4.12e-82
$\hat{\rho}$	-0.007923	Durbin-Watson		2.008208

F-tests of zero restrictions

All lags of gdp	$F(1, 395) = 45.2359$	[0.0000]
All lags of infl	$F(1, 395) = 69.3755$	[0.0000]
All lags of rate	$F(1, 395) = 174.409$	[0.0000]
All vars, lag 1	$F(3, 395) = 212.767$	[0.0000]

B.2 Impulse Responses to a Unit Standard Error Shock in Variables

Responses to a one-standard error shock in gdp

period	gdp	infl	rate
1	0.456313	0.114911	0.276116
2	0.167512	0.172359	0.260130
3	-0.00762407	0.128074	0.213430
4	-0.0831068	0.0602156	0.142562
5	-0.0914028	0.00645359	0.0721134
6	-0.0665558	-0.0223372	0.0193915
7	-0.0340809	-0.0297198	-0.0104446
8	-0.00810966	-0.0243712	-0.0209695
9	0.00675936	-0.0144817	-0.0193820
10	0.0117825	-0.00540176	-0.0126373
11	0.0105825	0.000544006	-0.00554671
12	0.00678726	0.00319559	-0.000474131
13	0.00291540	0.00347717	0.00208464
14	0.000187695	0.00254877	0.00266572
15	-0.00116465	0.00132967	0.00212904
16	-0.00145134	0.000344798	0.00122481
17	-0.00114453	-0.000228196	0.000421773
18	-0.000649539	-0.000430693	-8.82321e-005
19	-0.000216246	-0.000395434	-0.000303478
20	5.55139e-005	-0.000258021	-0.000313074
21	0.000167630	-0.000114027	-0.000222087
22	0.000169718	-1.08493e-005	-0.000111103
23	0.000118917	4.14591e-005	-2.44918e-005
24	5.85026e-005	5.36476e-005	2.41130e-005

Responses to a one-standard error shock in infl

period	gdp	infl	rate
--------	-----	------	------

1	0.000000	0.381720	0.0183867
2	-0.117842	0.202156	0.126843
3	-0.150301	0.0731898	0.101425
4	-0.124935	-0.00304280	0.0428031
5	-0.0768393	-0.0353169	-0.00446617
6	-0.0313978	-0.0386415	-0.0281505
7	-0.000582546	-0.0279283	-0.0318685
8	0.0140877	-0.0141139	-0.0243421
9	0.0166977	-0.00316316	-0.0135163
10	0.0128180	0.00303650	-0.00432274
11	0.00705430	0.00506788	0.00133288
12	0.00216052	0.00449167	0.00359988
13	-0.000824397	0.00284963	0.00357297
14	-0.00199097	0.00120117	0.00246783
15	-0.00193765	5.21661e-005	0.00118909
16	-0.00132113	-0.000509246	0.000218624
17	-0.000625227	-0.000620384	-0.000309092
18	-0.000103971	-0.000483814	-0.000465428
19	0.000175289	-0.000271303	-0.000397660
20	0.000254411	-8.75870e-005	-0.000244133
21	0.000214345	2.63883e-005	-9.67273e-005
22	0.000129907	7.24617e-005	2.79051e-006
23	5.01642e-005	7.22286e-005	4.91981e-005
24	-3.01313e-006	5.00762e-005	5.62226e-005

Responses to a one-standard error shock in rate

period	gdp	infl	rate
1	0.000000	0.000000	0.124908
2	-0.0236964	-0.00339041	0.0611688
3	-0.0237955	-0.00963207	0.0242638
4	-0.0149796	-0.0119699	0.00424474
5	-0.00556804	-0.0103715	-0.00455253
6	0.000867450	-0.00683287	-0.00652251
7	0.00376781	-0.00322460	-0.00513345
8	0.00403981	-0.000591274	-0.00277085
9	0.00295431	0.000813478	-0.000747620
10	0.00154467	0.00122162	0.000464066
11	0.000406889	0.00103828	0.000907142
12	-0.000256364	0.000632575	0.000844510
13	-0.000492596	0.000246141	0.000558596
14	-0.000454250	-1.27833e-005	0.000252984
15	-0.000297333	-0.000131964	3.09073e-005
16	-0.000132044	-0.000148339	-8.38808e-005
17	-1.32406e-005	-0.000110868	-0.000112752
18	4.72240e-005	-5.92479e-005	-9.20363e-005
19	6.15290e-005	-1.66570e-005	-5.41051e-005
20	4.95465e-005	8.65072e-006	-1.95779e-005
21	2.87365e-005	1.80286e-005	2.79436e-006
22	1.00848e-005	1.69800e-005	1.25664e-005
23	-1.85018e-006	1.13002e-005	1.33724e-005
24	-6.95332e-006	5.15444e-006	9.67438e-006

B.3 ARMA(2,1) Model for Inflation Rate

Model 8: ARMA, using observations 1973:01–2006:04 ($T = 400$)

Dependent variable: infl

Standard errors based on Hessian

	Coefficient	Std. Error	z	p-value
const	0.0510611	0.0318144	1.605	0.1085
ϕ_1	1.22779	0.138513	8.864	0.0000
ϕ_2	−0.489008	0.0731917	−6.681	0.0000
θ_1	−0.598430	0.150242	−3.983	0.0001
Mean dependent var	0.050592	S.D. dependent var	0.510783	
Mean of innovations	−0.000610	S.D. of innovations	0.413233	
R^2	0.343851	Adjusted R^2	0.340546	
Log-likelihood	−214.3227	Akaike criterion	438.6454	
Schwarz criterion	458.6028	Hannan–Quinn	446.5488	

			Real	Imaginary	Modulus	Frequency
AR						
	Root	1	1.2554	−0.6848	1.4300	−0.0795
	Root	2	1.2554	0.6848	1.4300	0.0795
MA						
	Root	1	1.6710	0.0000	1.6710	0.0000

Test for normality of residual –

Null hypothesis: error is normally distributed

Test statistic: $\chi^2(2) = 0.360754$

with p-value = 0.834955

LM test for autocorrelation up to order 12 –

Null hypothesis: no autocorrelation

Test statistic: $\chi^2(9) = 5.97123$

Test for ARCH of order 12 –

Null hypothesis: no ARCH effect is present

Test statistic: LM = 6.80325

with p-value = $P(\chi^2(12) > 6.80325) = 0.870337$

- Exercise 2 -

B.4 Ordinary Least Squares (OLS) performed on m with y and i as regressors

Model 10: OLS, using observations 1973:01–2006:04 ($T = 400$)

Dependent variable: m

	Coefficient	Std. Error	t -ratio	p-value
const	−0.951311	0.468009	−2.033	0.0428
y	1.00413	0.00460784	217.9	0.0000
i	−0.285938	0.00723542	−39.52	0.0000
Mean dependent var	89.19301	S.D. dependent var	41.84973	
Sum squared resid	5748.857	S.E. of regression	3.805358	
R^2	0.991773	Adjusted R^2	0.991732	
$F(2, 397)$	23930.39	P-value(F)	0.000000	
Log-likelihood	−1100.634	Akaike criterion	2207.268	
Schwarz criterion	2219.242	Hannan–Quinn	2212.010	
$\hat{\rho}$	0.645811	Durbin–Watson	0.708276	

B.5 Error Correction Model (ECM) of quantity of money (m)

Model 15: OLS, using observations 1973:03–2006:04 ($T = 398$)

Dependent variable: d_m

	Coefficient	Std. Error	t -ratio	p-value
const	0.388515	0.100272	3.875	0.0001
d_y_1	−0.00486263	0.0595422	−0.08167	0.9350
d_i_1	0.0252644	0.0252776	0.9995	0.3182
$resi_ols_m_1$	−0.369142	0.0310778	−11.88	0.0000
d_m_1	−0.00175892	0.0412460	−0.04264	0.9660
Mean dependent var	0.388264	S.D. dependent var	2.340692	
Sum squared resid	1436.750	S.E. of regression	1.912028	
R^2	0.339455	Adjusted R^2	0.332732	
$F(4, 393)$	50.49086	P-value(F)	2.76e−34	
Log-likelihood	−820.1912	Akaike criterion	1650.382	
Schwarz criterion	1670.315	Hannan–Quinn	1658.277	
$\hat{\rho}$	−0.002768	Durbin's h	−0.097179	

B.6 Vector Error Correction Model (VECM) for the First Difference of Variables

VECM system, lag order 1

Maximum likelihood estimates, observations 1973:02–2006:04 ($T = 399$)

Cointegration rank = 1

Case 3: Unrestricted constant

Cointegrating vectors (standard errors in parentheses)

m_{t-1}	1.00000 (0.000000)
y_{t-1}	-1.00283 (0.00642445)
i_{t-1}	0.284841 (0.0100609)

Adjustment vectors

m_{t-1}	1.00000
y_{t-1}	-0.0723359
i_{t-1}	-0.294878

Log-likelihood = -2762.22

Determinant of covariance matrix = 206.872

AIC = 13.9059

BIC = 14.0258

HQC = 13.9534

Equation 1: Δm

	Coefficient	Std. Error	t-ratio	p-value
const	0.0879518	0.0976589	0.9006	0.3683
EC1	-0.357516	0.0251196	-14.23	0.0000
Mean dependent var	0.388210	S.D. dependent var	2.337749	
Sum squared resid	1440.231	S.E. of regression	1.904674	
R^2	0.337855	Adjusted R^2	0.336187	
$\hat{\rho}$	-0.015706	Durbin-Watson	2.031339	

Equation 2: Δy

	Coefficient	Std. Error	t-ratio	p-value
const	0.442612	0.0942576	4.696	0.0000
EC1	0.0258613	0.0242447	1.067	0.2868
Mean dependent var	0.420892	S.D. dependent var	1.838657	
Sum squared resid	1341.658	S.E. of regression	1.838339	
R^2	0.002858	Adjusted R^2	0.000346	
$\hat{\rho}$	-0.009199	Durbin-Watson	2.014695	

Equation 3: Δi

	Coefficient	Std. Error	t-ratio	p-value
const	0.191725	0.212892	0.9006	0.3684
EC1	0.105424	0.0547594	1.925	0.0549
Mean dependent var	0.103185	S.D. dependent var	4.166191	
Sum squared resid	6844.247	S.E. of regression	4.152098	
R^2	0.009250	Adjusted R^2	0.006754	
$\hat{\rho}$	0.006463	Durbin-Watson	1.986884	

Cross-equation covariance matrix

	Δm	Δy	Δi
Δm	3.60960	0.0350345	-0.364633
Δy	0.0350345	3.36255	-0.491674
Δi	-0.364633	-0.491674	17.1535

determinant = 206.872